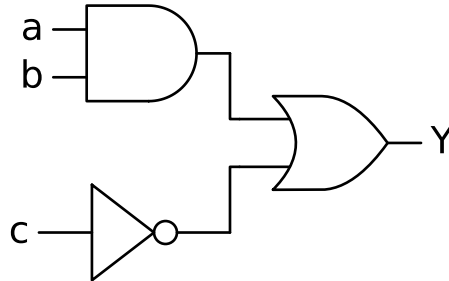


Boolean Algebra Worksheet

Answer key

Problem 1.



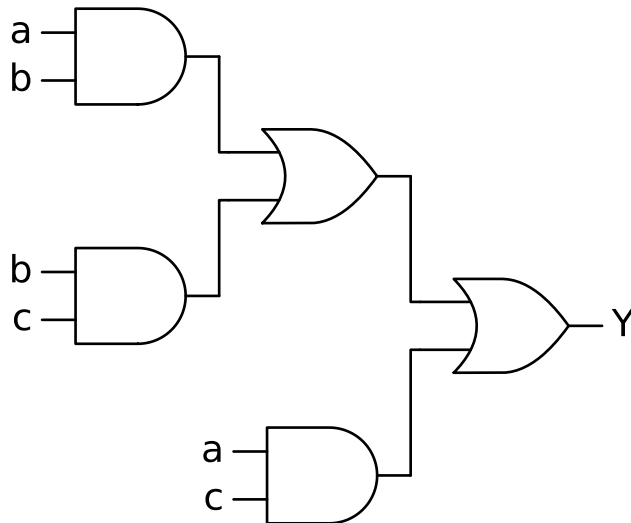
Expression: $A \cdot B + \overline{C}$

Simplifies to: $\overline{C} + A \cdot B$

Truth table:

A	B	C	Y
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

Problem 2.



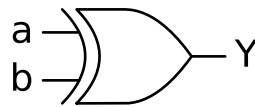
Expression: $A \cdot B + B \cdot C + A \cdot C$

Simplifies to: $A \cdot B + A \cdot C + B \cdot C$

Truth table:

A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Problem 3.



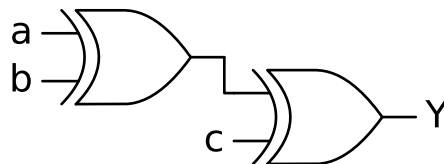
Expression: $A \oplus B$

Simplifies to: $A \cdot \overline{B} + B \cdot \overline{A}$

Truth table:

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Problem 4.



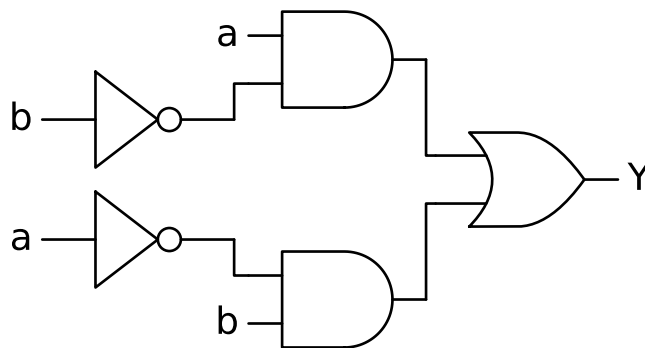
Expression: $A \oplus B \oplus C$

Simplifies to: $A \cdot B \cdot C + A \cdot \overline{B} \cdot \overline{C} + B \cdot \overline{A} \cdot \overline{C} + C \cdot \overline{A} \cdot \overline{B}$

Truth table:

A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

Problem 5.

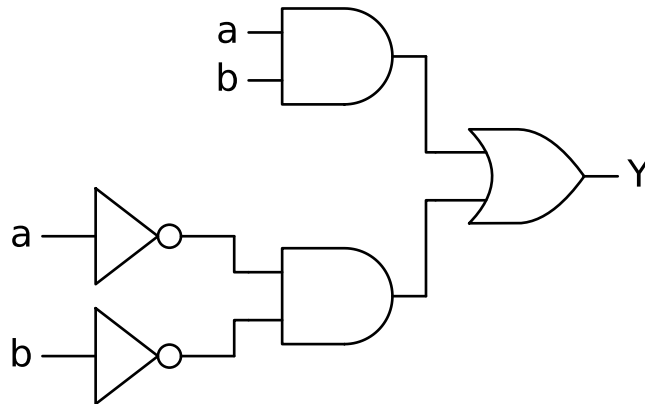


Expression: $A \cdot \overline{B} + \overline{A} \cdot B$
Simplifies to: $A \cdot \overline{B} + B \cdot \overline{A}$

Truth table:

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Problem 6.

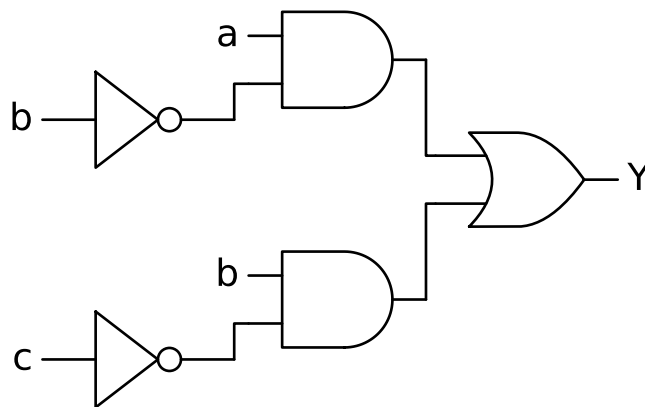


Expression: $A \cdot B + \overline{A} \cdot \overline{B}$

Truth table:

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

Problem 7.

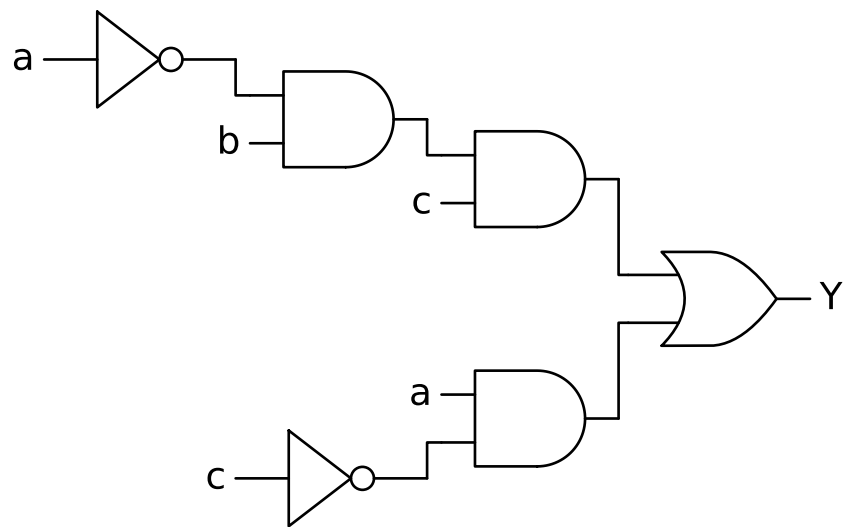


Expression: $A \cdot \overline{B} + B \cdot \overline{C}$

Truth table:

A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Problem 8.



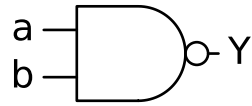
Expression: $\bar{A} \cdot B \cdot C + A \cdot \bar{C}$

Simplifies to: $A \cdot \bar{C} + B \cdot C \cdot \bar{A}$

Truth table:

A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

Problem 9.



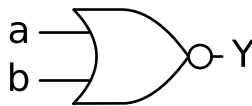
Expression: $\overline{A \cdot B}$

Simplifies to: $\overline{A} + \overline{B}$

Truth table:

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

Problem 10.



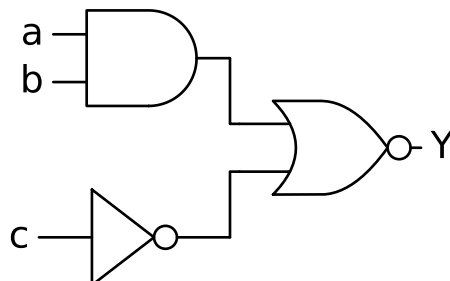
Expression: $\overline{A + B}$

Simplifies to: $\overline{A} \cdot \overline{B}$

Truth table:

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

Problem 11.



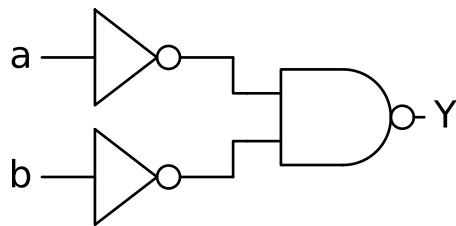
Expression: $\overline{A \cdot B + C}$

Simplifies to: $C \cdot \overline{A} + C \cdot \overline{B}$

Truth table:

A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	0

Problem 12.



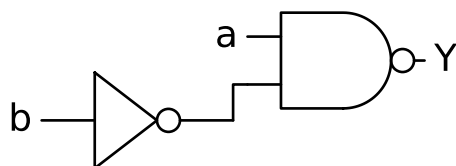
Expression: $\overline{\overline{A} \cdot \overline{B}}$

Simplifies to: $A + B$

Truth table:

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

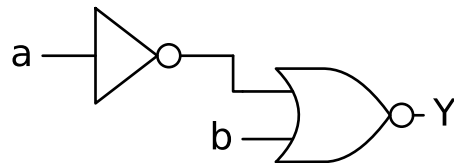
Problem 13.



Expression: $\overline{A \cdot \overline{B}}$
Simplifies to: $B + \overline{A}$
Truth table:

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	1

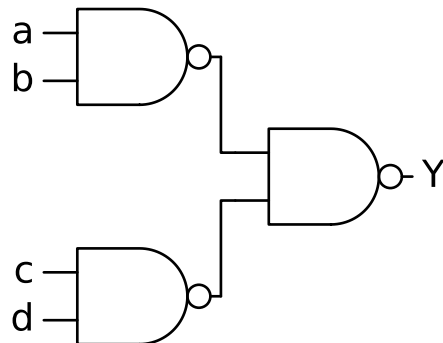
Problem 14.



Expression: $\overline{\overline{A} + B}$
Simplifies to: $A \cdot \overline{B}$
Truth table:

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	0

Problem 15.



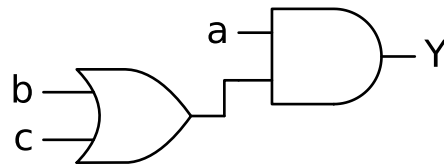
Expression: $\overline{\overline{A \cdot B \cdot C \cdot D}}$
Simplifies to: $A \cdot B + C \cdot D$
Truth table:

A	B	C	D	Y
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

Problem 16.

Draw the logic-gate circuit for:

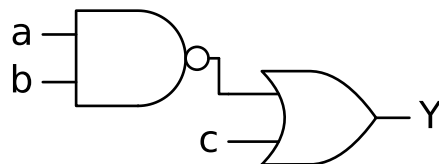
$$Y = A \cdot (B + C)$$



Problem 17.

Draw the logic-gate circuit for:

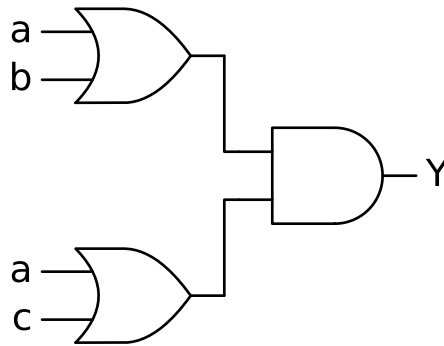
$$Y = \overline{A \cdot B} + C$$



Problem 18.

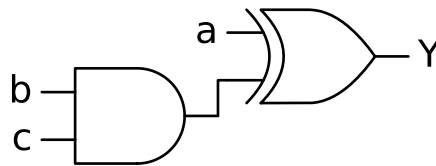
Draw the logic-gate circuit for:

$$Y = (A + B) \cdot (A + C)$$

**Problem 19.**

Draw the logic-gate circuit for:

$$Y = A \oplus B \cdot C$$

**Problem 20.**

Draw the truth table for:

$$Y = A \cdot B + C$$

A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Problem 21.

Draw the truth table for:

$$Y = A \oplus B \oplus C$$

A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

Problem 22.

Draw the truth table for:

$$Y = \overline{A + B} \cdot C$$

A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

Problem 23.

Draw the truth table for:

$$Y = A \cdot \overline{B} + \overline{A} \cdot C$$

A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0
